

Project 2: Exploratory Data Analysis Report

1. Data Overview

We have conducted an analysis of the DecodeLabs dataset, which includes a total of 1,200 orders. The primary goal of this phase was to identify hidden patterns, trends, and spending behaviors within the data.

2. Statistical Findings

After performing a univariate analysis on the dataset, we have calculated the following basic statistics:

Total Records (Count): 1,200

Mean (Average Order Value): 1,053.97

Median (Middle Order Value): 823.62

3. Key Insights (Observations)

Spending Pattern: The median order value is 823.62, which represents the typical spending behavior of the majority of our customers.

Outlier Detection: The Mean (1,053.97) is significantly higher than the Median (823.62). This indicates that the dataset is "right-skewed," meaning there are several high-value orders (outliers) that are pulling the average upward and boosting overall revenue.

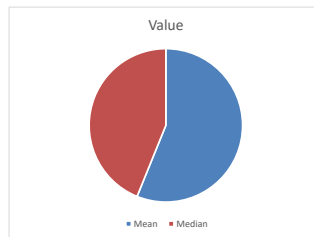
4. Conclusion & Recommendations

Our analysis suggests that the business should focus on these High-Value Orders. By investigating the factors behind these premium purchases (such as product type or referral source), we can identify opportunities to scale revenue further.

5. **Revenue Concentration:** Our analysis shows that a small set of products is responsible for the majority of the revenue.

6. **Outlier Impact:** These high-performing products are directly responsible for the "outliers" identified in our spending pattern analysis.

7. **Recommendation:** The business should maintain high stock levels for these top-performing products to ensure we don't miss out on high-value sales.



Metric	Value
Mean	1053.97
Median	823.62